



Zwitterionic–polyurethane coatings for non-specific marine bacterial inhibition: A nontoxic approach for marine application

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ABSTRACT

The present work reports synthesis and development of zwitterionic tri-block copolymer-polyurethane (ZTC-PU) coatings which are thermally stable and have hydrophobic properties which can be used for marine applications. The zwitterionic tri-block copolymer (ZTC) (poly(sulfobetaine methacrylate) - polydimethylsiloxane - poly(sulfobetaine methacrylate), polySBMA-block-PDMS-block-polySBMA) was synthesized by ATRP (atom transfer radical polymerization) method. Apart from this, the facile synthesis of silane functionalized acrylic polyol was formulated with 4,4'-Methylene dicyclohexyl diisocyanate (H12 MDI) for the development of polyurethane (PU) coatings. Different weight ratios of ZTC was incorporated in PU to obtain a zwitterionic tri-block copolymer-polyurethane (ZTC-PU) coatings. The ZTC-PU coating exhibit good thermal stability and hydrophobic property. The coating showed excellent non-specific marine bacterial inhibition property along with good stability against gram positive *Staphylococcus aureus* (*S. aureus*) and gram negative *Escherichia coli* (*E. coli*), *Pseudomonas aeruginosa* (*P. aeruginosa*) microbial strains. This design method reveals the development of zwitterionic-polyurethane coatings with higher amount of zwitterionic moiety in the coating system, that will be efficient and eco-friendly technology for marine bio-fouling application.

1. Introduction

In general, bio-fouling or fouling is formed due to the accumulation and growth of marine organisms (algae, barnacles, mollusks) on the surface immersed in the marine water [1]. Different species of fouling organisms are found in different regions of the sea. They get spread from one region to another through ship transportation. The fouled ship hull decreases the ship mobility; increases drag friction and fuel consumption [2–4]. Tributyltin (TBT) and copper based anti-fouling paints effectively combat fouling organism. The flip side of these coating systems is persistent toxicity in non-targeted aquatic organism such as fish and marine mammals of porpoise and grey seals. Use of tin based coatings was banned by international maritime organisation (IMO) in January 2003 [5]. In recent times, the marine industries and researchers are focusing on a non-toxic anti-fouling coating such as Polyethylene glycol (PEG) polymers [6], fluoropolymers [7], xerogels [8] silicone elastomers [9] and zwitterionic polymers [10,11]. The ability to attach and settle fouling organism on coatings depend on surface and bulk properties of materials such as surface free energy, wettability, roughness and elastic modulus [12–15]. To develop such a coating having of all the above mentioned properties requires a high degree of complexity.

In the current scenario, use of zwitterionic polyurethane coating provides foremost approach to eliminate the aforesaid

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complexities. In addition, zwitterionic polyurethane coatings are also considered as potential non-toxic anti-fouling and foul release coatings as reported by Bodkhe et al., [16]. Further, the surface and bulk properties of the coating can be fine-tuned by introducing amphiphilic character by adjusting hydrophilic to hydrophobic unit ratio in the coating system. In this regard, zwitterionic polymers have been used as hydrophilic polymer, also this polymer was found as an environmentally benign ultra-low marine bio-fouling candidate [17–19]. Initially, the zwitterionic polymers were used in the biomedical applications and subsequently they have found their application in marine fouling [20,21]. Owing to their high degree of solvation in water, strong electrostatic induced hydration, tightly bound water layer in the vicinity of polymer chain aids to prevent the attachment of non-specific protein as well as the adhesion of various types of cells [22]. There are several report investigated the non-fouling properties of Poly(Sulfobetaine Methacrylate) (polySBMA) grafted polypropylene [23]. Polyethylene terephthalate [24], polystyrene and polyurethane [25–27]. It was reported that the coatings were densely packed in most of the published reports.

The silicone elastomers with low surface energy and low modulus are preferred choices for reduction of densely packed system as well as excellent for foul releasing properties. Typically, polydimethylsiloxane (PDMS) is the most commonly used model for low surface energy material in laboratory, and increasingly finding its application in marine environment [28]. In general, low surface energy material has feeble interaction with marine organism. Therefore, accumulated marine organism can be easily removed by mechanical cleaning, water spray or dragging force [29]. In this regard self-stratified siloxane-polyurethane foul release coating has been extensively reported by D.C. Webster group [30–32]. These coatings suffer to preserve attachment of diverse organisms present in the marine. Example, *Ulva linza* (*U. linza*) adheres weakly to hydrophobic surface whereas *Navicula incerta* (*N. Incerta*) adheres strongly to hydrophobic surfaces. Therefore, the choice of designing a coating with combination of hydrophilic and hydrophobic component can cover and control the broad spectrum of bio-fouling species present in the marine environment [33]. To design an amphiphilic coating, polyethylene glycol is the preferred material for hydrophilic counterpart due to its non-toxic nature and non specific protein resistant characteristics [34]. The protein and cell resistance of the PEG based materials includes large exclusion volume, high mobility, strong hydration force and steric hindrance. However, polyethylene glycol undergoes autooxidation in the marine environments [35]. As a result, there is a change in the chemical structure which decreases the fouling inhibition properties that is not useful for long term applications. In the recent years, development of amphiphilic coating with zwitterionic materials are gaining considerable research interests, Mostly sulfobetaine, carboxybetaine and phosphobetaine materials are preferred due to high affinity of binding water molecule and strong electrostatically induced hydration. These materials have already been investigated as stable and ultra-low fouling materials. Among them sulfobetaine methacrylate (SBMA) is used in a wide range of applications due to its biocompatible nature and high resistant to protein, bacterial adsorption and substantial reduction in the bio film formation [36].

In this article, a series of zwitterionic triblock copolymer-polyurethane (ZTC-PU) coatings ZTC-PU 10%, ZTC-PU 20% and ZTC-PU 30% were prepared. Formerly, bi-functional PDMS ATRP macro initiator was synthesised according to the report [37]. In this study, we have used PDMS whose Mn is 3000. Further, a macro initiator was used to polymerize the SBMA to obtain a PolySBMA-block-PDMS-block-polySBMA triblock copolymer with high molecular weight by ATRP (atom transfer radical polymerization). Simultaneously, silane functionalized acrylic polyol were synthesized. The silane functionalized acrylic polyol, 4,4'-Methylene dicyclohexyl diisocyanate (H12 MDI) and different weight percentage (10, 20, 30 wt%) of zwitterionic triblock copolymer was formulated to obtain ZTC-PU coatings. In the coating formulation 1,4-butanediol was used as chain extender. Thermal, mechanical and surface morphologies of the coatings were investigated. Gram positive and negative bacterial assessment and nonspecific marine bacterial inhibition was also studied.

2. Experimental

2.1. Materials

2-Bromoisobutyl bromide (BIBB, 98%), 2-Hydroxyethyl acrylate (HEA), 2,2'-bipyridine (bpy 99%), copper (I) bromide ($\text{Cu}^{\text{I}}\text{Br}$), triethylamine 99%, Anhydrous THF, N-methyl-2-pyrrolidone (NMP), Acetyl acetone, 2-propanol and methanol were purchased from TCI chemicals, Chennai, India. Bis(3-aminopropyl)-terminated polydimethylsiloxane, $M_n = 3000$ (APT-PDMS, 3000) was received from Gelest, PA, USA. Acrylic polyol (Duracryl-7865) and N-benzyl-N-aminoethyl-3-aminopropyltrimethoxy silane was purchased from D. R. Coats and Evonik, Mumbai, India. [2-(Methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl)ammonium hydroxide (SBMA 97%), 1,4-butanediol, 4,4'-Methylene dicyclohexyl diisocyanate (H12 MDI), Dibutyltin dilaurate (DBTL), were obtained from Sigma Aldrich, Missouri, USA. All the reagents and solvents were used as received without any further purification.

2.2. Characterization techniques

The ^1H NMR spectra for all the synthesized samples were recorded using Varion-Inova-500 MHz spectrometer taking tetramethyl silane (TMS) as the internal standard at ambient temperature in CDCl_3 and D_2O solvents. Fourier transform infrared spectroscopy (FTIR) analysis were recorded on a Thermo scientific, Nicolet 6700, USA, with a attenuated total reflection (ATR) mode. The wavenumber ranges from 400 to 4000 cm^{-1} with total scan of 32. Thermogravimetric analysis (TGA) TGA Q50, TA Instrument USA was used to study the thermal stability of the coatings. A heating rate of $10\text{ }^\circ\text{C}/\text{min}$ with continuous N_2 flow of 30 ml/min. Differential scanning calorimetry (DSC) was performed for all the coatings using thermal analyzer DSC, Q20, TA instruments, USA. A heating rate of $10\text{ }^\circ\text{C}/\text{min}$ from $-50\text{ }^\circ\text{C}$ to $150\text{ }^\circ\text{C}$ was employed. The viscosity (η) of the zwitterionic triblockcopolymer was examined using modular advanced rheometer MARS III, equipped with fluid bath temperature control and couette (diameter 25 mm) Thermo Fisher Scientific, Germany. Surface morphology of the coatings was analyzed by scanning electron microscopy (SEM) EVO MA 15, Carl Zeiss

SMT Ltd, UK. The thin film samples were coated with mixture of gold and palladium using vacuum sputter prior to analysis to improve the conductivity of the surface. Water contact angles were measured using Holmarc HO-IAD-CAM-01 instrument.

2.2.1. Bacterial assay

The anti-bacterial activity of the coating materials were studied with standard strains [(Gram positive and Gram negative) obtained from microbial type culture collection centre, IMTECH, Chandigarh], as well as against nonspecific marine bacteria isolated from Bay of Bengal.

2.2.1.1. Antibacterial activity study of ZTC-PU films against specific marine bacteria. Gram positive bacteria *Staphylococcus aureus* (MTCC-96) and gram negative bacteria *Escherichia coli* (MTCC 443), *Pseudomonas aeruginosa* (MTCC 741) were procured from Microbial type Culture Collection Centre (MTCC), Institute of Microbial Technology (IMTECH), Chandigarh, India and used in the study. Nutrient Agar (Hi-Media Ltd, Mumbai) plates were prepared as per the manufacturer's instructions. Overnight fresh cultures of the strains were swept over Nutrient agar (NA) plates separately using a sterile cotton swab.

The ZTC-PU films were cut into appropriate dimension and washed with sterile distilled water and applied over the sweep culture. The plates were incubated at $35 \pm 2^\circ\text{C}$ for 18–24 h and observed for zones of inhibition around the films. The experiments were carried out in triplicate. A zone of inhibition around the films indicates the positive antibacterial activity of the respective film.

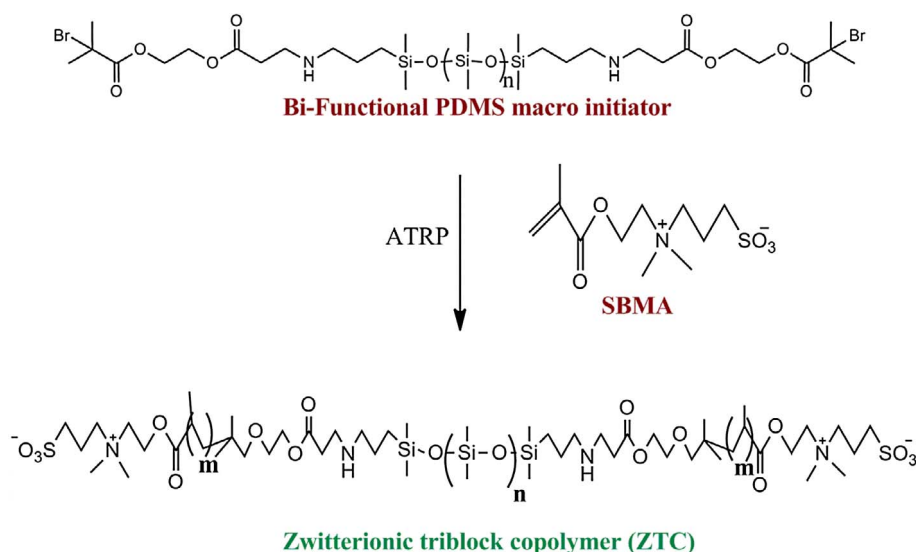
2.2.1.2. Antibacterial activity study of ZTC-PU films against non-specific marine bacteria. Marine bacterial strains were isolated from sea water collected from Bay of Bengal, India at Puri (Latitude: $19^\circ47'53''\text{N}$, Longitude: $85^\circ49'29''\text{E}$). Briefly NA plates were prepared with 50% aged sea water, as described previously [38]. One loop full of sample (3–5 μl approximately) was streaked onto the above nutrient agar plates. On the other hand 50 and 100 μl of water samples were applied on to the surface of nutrient agar plates separately and spread by the help of a sterile glass rod. The plates were incubated at $35 \pm 2^\circ\text{C}$ for overnight and observed for presence of bacterial growth. Selected colonies (based on colony morphology and growth characters) were pure cultured and preserved for future use. The antibacterial activity of the ZTC-PU films against these isolates were tested as described earlier.

2.3. Reaction scheme

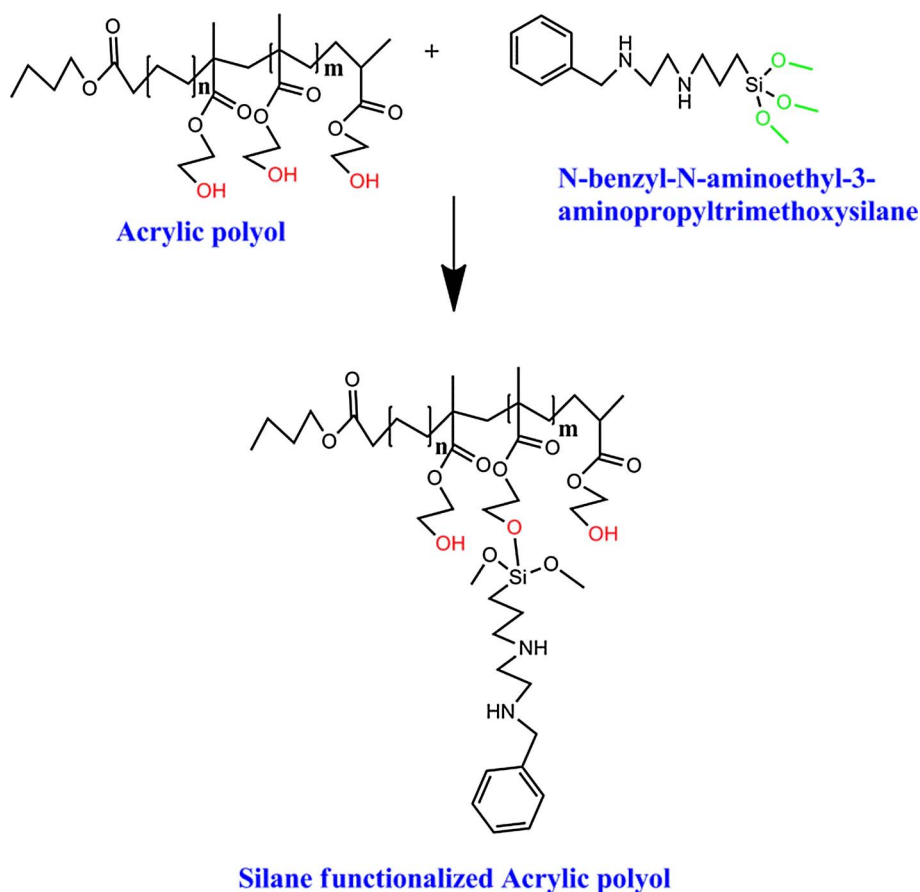
The synthesis route for polySBMA-block-PDMS-block-polySBMA triblock copolymer (Scheme 1). Synthesis of silane functionalized acrylic polyol was given in the Scheme 2, and formation of ZTC-PU coatings represented in Scheme 3.

2.4. Synthesis of polySBMA-PDMS-polySBMA triblock copolymer using ATRP method

Polymerization of SBMA was carried out by using bifunctional PDMS macroinitiator. PDMS macroinitiator was synthesized as similar method used in the report [37]. PDMS macroinitiator (3.49 g, 0.1×10^{-2} mol) in 10 ml methanol and 2.5 ml 2-propanol solvent was taken in a 250 ml two neck round bottom flask equipped with nitrogen inlet and magnetic stirrer. After purging with nitrogen for 30 min, Cu(I)Br catalyst (0.145 g, 0.1×10^{-2} mol) and bpy ligand (0.3122 g, 0.2×10^{-2} mol) were added to the solution which was then stirred for 30 min to form a complex under nitrogen atmosphere. SBMA (5.0 g, 1.78×10^{-2} mol) was added to the reaction solution, which attained a light green colour. The reaction was continued for 48 h to complete the polymerization



Scheme 1. Synthesis route of PolySBMA-block-PDMS-block-polySBMA triblock copolymer.



Scheme 2. Synthesis of silane functionalized acrylic polyol.

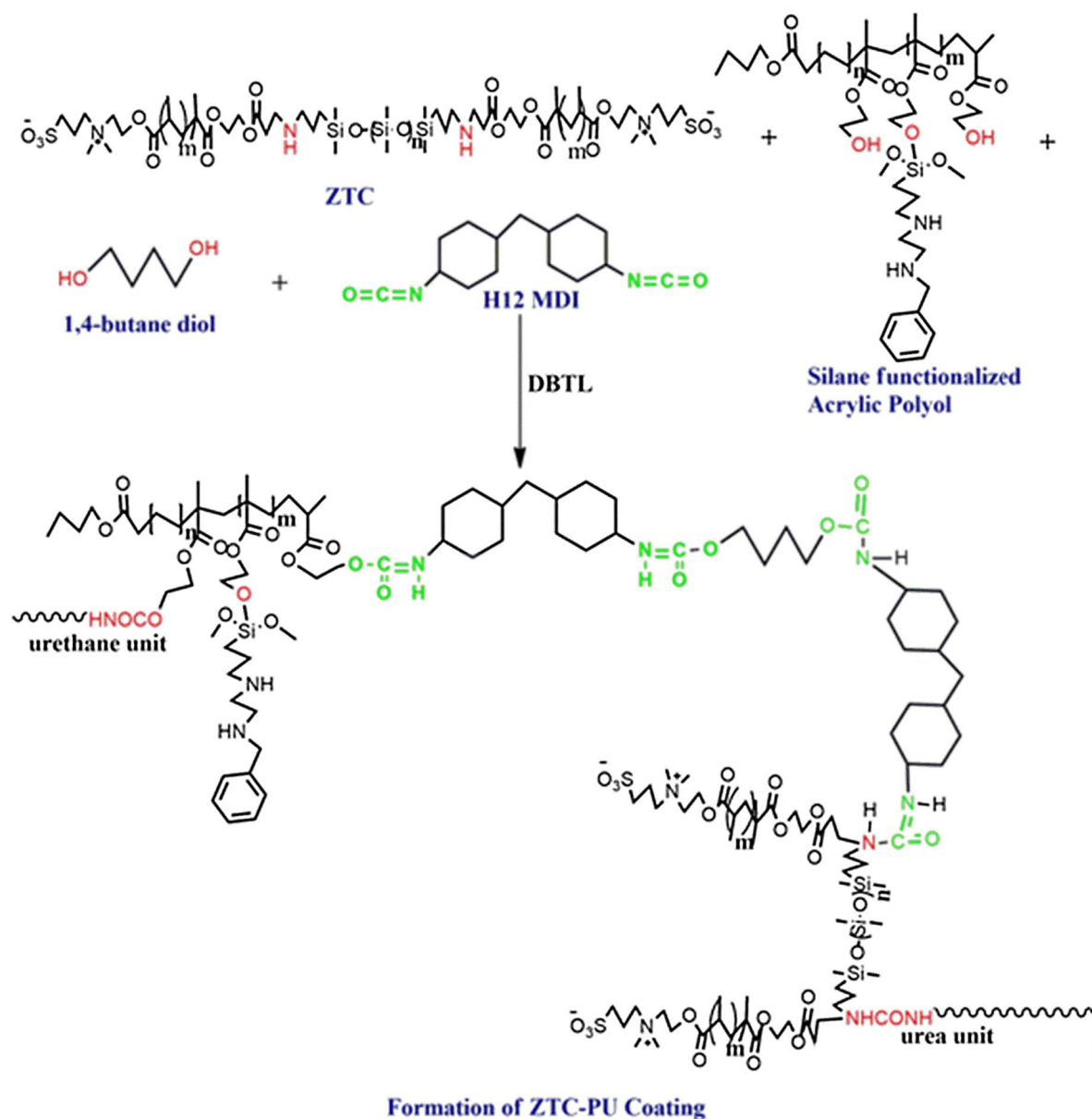
process (Scheme 1). Subsequently, ^1H NMR was taken and the result suggested the disappearance of vinyl peak at δ 5.5–6.0 confirms the SBMA has been polymerized. In the ATRP reaction $[\text{SBMA}]:[\text{Br-PDMS-Br}]:[\text{CuBr}]:[\text{Bpy}] = 100:1:1:2$ was maintained. The polymeric solution was passed through a short silica column (silica gel 100–200 mesh) to remove the spent catalyst. The colourless solution was precipitated into THF. The triblock copolymer was dried in vacuum oven at room temperature for at least 24 h. 10–20% of compound loss was observed during the purification due to adsorption of zwitterionic polymer onto the silica gel. The reaction scheme for the synthesis of the triblock copolymer is shown in Scheme 1. ^1H NMR (500 Mz, D_2O), δ (ppm), δ 1.3 (d, $\text{H}_3\text{C}-$), δ 1.81 (d, $-\text{CH}_2-$), δ 2.1 (m, $\text{CH}_2-\text{CH}_2\text{CH}_2\text{SO}_3^-$), δ 2.83 (t, $-\text{CH}_2\text{SO}_3^-$), δ 3.02 (m, CH_3-N), δ 3.38 (m, CH_2-N), δ 3.68 (s, CH_3-C), δ 3.9 (t, CH_2-CO) and disappearance of acrylic peak at δ 5.5–6.5. The viscosity (η) 6610 mPa s at 25 °C of the zwitterionic triblock copolymer was analyzed using rheometer.

2.5. Synthesis of amino silane functionalized acrylic polyol

Acrylic polyol(Duracryl-7861)- 10 g, 0.02 mol, N-benzyl-N-aminoethyl-3-aminopropyltrimethoxy silane (Dynasylan-8461)- 1.53 g, 0.004 mol and anhydrous THF 20 ml were taken in a two neck round bottom flask equipped with water condenser, nitrogen inlet and magnetic stirrer. The reaction mixture was heated to 80 °C for 8 h under nitrogen atmosphere. After that, solvent was removed by azeotropic condensation method, the product was extracted in chloroform and washed with ethanol 2–3 times to remove unreacted aminosilane. Eventually the solvent was evaporated to obtain aminosilane functionalized acrylic polyol (Scheme 2). The product was further characterized by FT-IR and ^1H NMR. ^1H NMR (500Mz CDCl_3), δ (ppm), δ 0.82 ($-\text{Si}-\text{CH}_2-$), δ 0.92 ($-\text{CH}_3$), δ 1.20 ($-\text{CH}_2-$), δ 1.32 ($-\text{NHCH}_2$), δ 1.55 ($-\text{NHCH}_2\text{Ar}$), δ 2.30 ($-\text{CH}_2\text{Ar}$), δ 2.64 ($\text{OH}-\text{CH}_2-$), δ 3.46 ($-\text{Si}-\text{OCH}_3$), δ 3.54 ($-\text{CH}_2-\text{COO}-$), δ 3.72 ($-\text{CH}_2-\text{OCO}-$), δ 7.05–7.18 ($-\text{C}_6\text{H}_5$ protons from silane moiety).

2.6. Synthesis of zwitterionic triblockcopolymer-polyurethane coatings (ZTC-PU)

Zwitterionic triblockcopolymer-polyurethane coating was formulated with different weight percentage (10, 20, 30) of zwitterionic triblock copolymer into the acrylic based polyurethane system. In 10 wt% formulation, 0.25 g of zwitterionic triblock copolymer and 0.37 g of N-methylpyrrolidone (NMP) (1:1.5 ratio) was taken in a 250 ml round bottom flask equipped with magnetic



Scheme 3. Formation of ZTC-PU coating.

stirrer, water condenser, nitrogen inlet. The mixture was stirred overnight at 60 °C. 2.29 g silane grafted acrylic polyol, 0.23 g 1,4-butanediol, 1.2 g H12 – MDI, 0.40 g Acetylacetone, 0.002 g DBTL was added into the reaction mixture and the homogeneous solution was stirred for 2 h at 65 °C to obtain a proper viscosity for the coating application. The solution was casted on Teflon sheet and allowed to cure for room temperature at laboratory humidity condition for at least 15 days to get a free film (Scheme 3). The formulation was also coated on epoxy primed mild steel panel (low carbon content 0.05%–0.25%,) for marine microbial testing. The same procedure was followed for the 20 and 30 wt% formulation with pre-calculated amount of zwitterionic triblock copolymer with respect to polyol to obtain zwitterionic triblockcopolymer-polyurethane coatings.

3. Result and discussion

3.1. PolySBMA-block-PDMS-block-polySBMA triblockcopolymer

Bis(3-aminopropyl)-terminated polydimethylsiloxane, $M_n = 3000$ (APT-PDMS, 3000) was used as a precursor for the synthesis bi-functional PDMS ATRP macroinitiator. The polySBMA-block-PDMS-block-polySBMA triblock copolymer was synthesized through ATRP reaction using the bi-functional macro initiator (Scheme 1). The triblock copolymer was confirmed by ¹H NMR and FT-IR

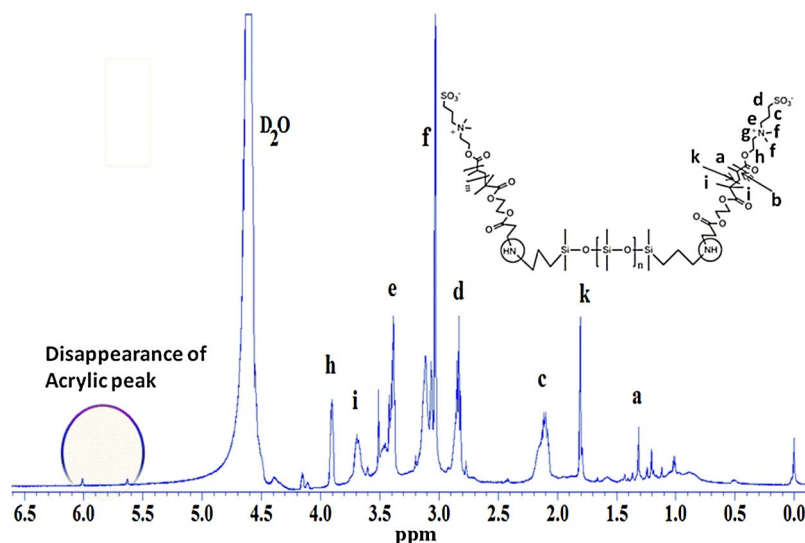


Fig. 1. ^1H NMR spectra of polySBMA-PDMS-polySBMA triblock copolymer.

respectively, ^1H NMR result (Fig. 1) shows the formation of zwitterionic triblock copolymer. Disappearance of acrylic peak at 5.5–6.5 confirms the SBMA monomers are polymerized. Along with characteristic chemical shift values at δ 1.3 (d, $\text{CH}_3\text{—CH}$), δ 1.81 (d, $\text{CH}_2\text{—CH}$), δ 2.1 (m, $\text{—CH}_2\text{—CH}_2\text{CH}_2\text{SO}_3$), δ 2.83 (t, $\text{—CH}_2\text{SO}_3$), δ 3.02 (m, $\text{CH}_3\text{—NCH}_3$), δ 3.38 (m, $\text{CH}_2\text{—CH}_2\text{N}$), δ 3.68 (s, $\text{CH}_3\text{—C}$), δ 3.9 (t, $\text{CH}_2\text{—CO}$), confirms the zwitterionic triblock copolymer was formed. In FT-IR spectra (Fig. 2) the observed absorption —NH stretching frequency at 3412 cm^{-1} , asymmetric —CH_2 , —CH_3 (2962 cm^{-1}), C=O str from initiator (1716 cm^{-1}), C=O str from SBMA (1636 cm^{-1}), —CH_2 & CH_3 bend (1456 cm^{-1} and 1319 cm^{-1}), Si—C str (1296 cm^{-1}), Si—O (1160 cm^{-1}), SO_3 str (1035 cm^{-1}), C—O str (954 cm^{-1}), displacement associated with motion of two Si atom (792 cm^{-1} , 600 cm^{-1}) these peaks are collectively attributed to the formation of PolySBMA-block-PDMS-block-polySBMA triblock copolymer [15,37,39].

3.2. Silane functionalized acrylic polyol

The silane functionalized acrylic polyol (Scheme 2) was confirmed by ^1H NMR and FT-IR spectroscopy. In ^1H NMR spectroscopy (Fig. 3) presence of peak at δ 0.87 (—Si—CH_2), δ 0.92 (terminal —CH_3), δ 1.19 (—CH_2 in the chain), δ 1.32 ($\text{—NHCH}_2\text{—}$), δ 1.54 ($\text{—NH—CH}_2\text{Ar}$), δ 3.46 (—Si—OCH_3), δ 7.05–7.18 (d, 2H & m, 3H of $\text{—C}_6\text{H}_5$) justifies the formation of silane functionalized acrylic polyol [40]. Also, in FT-IR spectroscopy (Fig. 4). The major peak observed at 3440 cm^{-1} corresponds to —OH , —NH stretching presence in polyol. Doublet peak at 2872 cm^{-1} and 2932 cm^{-1} showed the —CH asymmetric and symmetric stretching, carbonyl stretching at 1726 cm^{-1} (C=O) represents the ester group. Further the band observed at $1430\text{—}1500\text{ cm}^{-1}$ corresponds to phenyl ring present in the silane moiety. The combination of peaks at 1163 cm^{-1} and 1118 cm^{-1} corresponds to C—O and Si—O which confirmed the formation of silane grafted acrylic polyol [41].

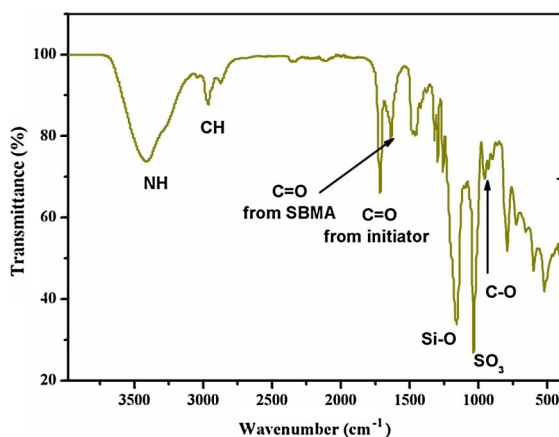


Fig. 2. FT-IR of polySBMA-PDMS-polySBMA triblock copolymer.

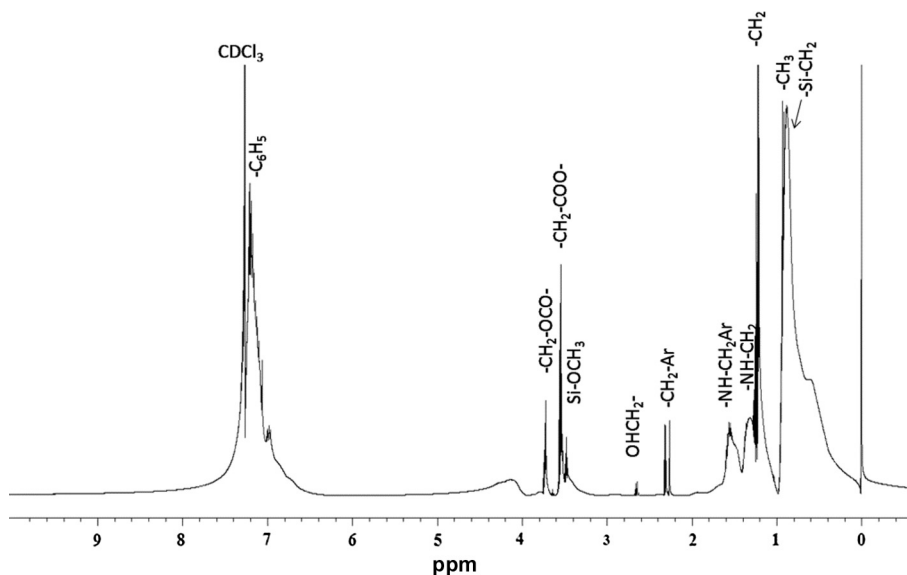
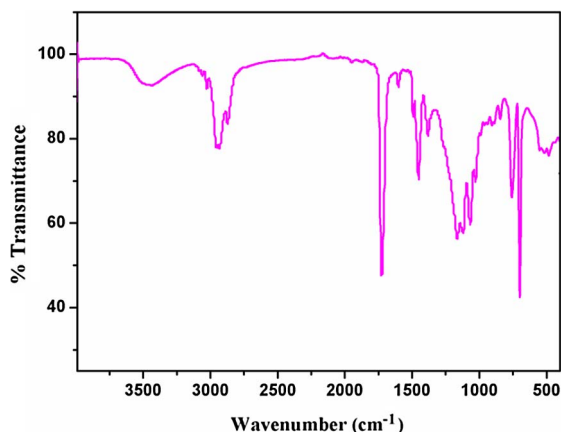
Fig. 3. ^1H NMR spectra of silane functionalized acrylic polyol.

Fig. 4. FT-IR spectra of silane functionalized acrylic polyol.

3.3. Zwitterionic triblockcopolymer-polyurethane(ZTC-PU) coatings

The reaction between triblock copolymer (ZTC), silane grafted acrylic polyol and H12 MDI produced a stable zwitterionic triblockcopolymer-polyurethane (ZTC-PU) coating. The formation of ZTC-PU coatings was confirmed by FT-IR spectroscopy (Fig. 5). Disappearance of $-\text{NCO}$ peak at around 2270 cm^{-1} confirms the complete moisture curing of coating. The series of observed peaks corresponds to urethane linkage, the peak at 3341 cm^{-1} corresponds to $-\text{NH}$ stretching in the urethane, the doublet of observed peak at 2850 and 2921 cm^{-1} is possibly related to $-\text{CH}$ symmetric/asymmetric stretch. The peak corresponds to 2850 cm^{-1} (symmetric $-\text{CH}_3$ stretch) and 2921 cm^{-1} (asymmetric $-\text{CH}_2$ stretch) respectively, amide - I consist of $-\text{C}=\text{O}$ stretching at 1718 cm^{-1} . combination of bands at 1527 cm^{-1} and 1448 cm^{-1} represent the amide-II region consisting of ($\delta_{\text{N-H}}$, $\nu_{\text{C-N}}$, $\nu_{\text{C-C}}$) stretch. $-\text{CH}_2$ symmetric and asymmetric stretching was observed at 1390 cm^{-1} . Amide -III region $\nu_{\text{C-N}}$, $\text{C}-\text{C}_{\alpha}$, $-\text{NH}$ bending was observed at $1190\text{--}1285\text{ cm}^{-1}$. The peak at 760 cm^{-1} and 700 cm^{-1} corresponds to amide IV $-\text{NH}$ out of plane and $-\text{CH}_2$ rocking respectively, which clearly shows the formation of polyurethane linkage [42]. Furthermore, appearance of peak at 1153 cm^{-1} associated with $-\text{Si}-\text{O}$ stretch and $-\text{SO}_3$ stretching from zwitterionic moiety appeared at 1039 cm^{-1} which merged with peak 1086 cm^{-1} corresponds to $-\text{CH}$ in plane bending. Also the observed peak at 801 cm^{-1} corresponds to motion displacement of two Si atoms [43]. With the increase in the content of zwitterionic triblockcopolymer from 10 to 30% within the polyurethane the intensity of $-\text{NH}$ region at 3341 cm^{-1} increased. This clearly supports the fact that more number of urethane-urea segment formation occurs in the coatings.

3.4. TGA and DSC analysis of ZTC-PU coatings

TGA (Thermo gravimetric analysis) results (Fig. 6, Table 1) show the thermal stability of zwitterionic triblockcopolymer-

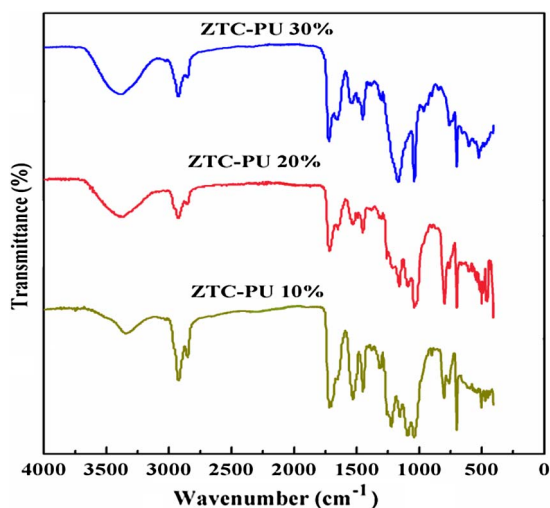


Fig. 5. FT-IR of zwitterionic triblockcopolymer-polyurethane (ZTC-PU).

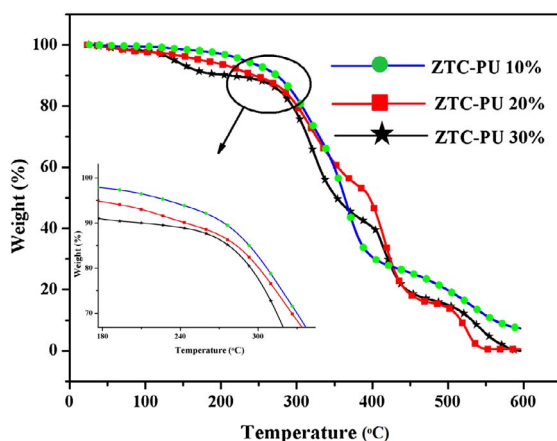


Fig. 6. TGA graph of zwitterionic triblockcopolymer-polyurethane (ZTC-PU).

Table 1

Thermo gravimetric analysis of zwitterionic triblockcopolymer-polyurethane (ZTC-PU).

Sample code	Onset decomposition temperature (T_{ON}) in °C	5% wt loss temperature (T_{d5}) in °C	20% wt loss temperature (T_{d20}) in °C	50% wt loss temperature (T_{d50}) in °C	% wt remaining at 400 °C
ZTC-PU10%	246	138	294	353	40
ZTC-PU20%	262	174	300	396	48
ZTC-PU30%	286	230	306	363	30

polyurethane (ZTC-PU) coatings. The onset degradation temperature (T_{ON}) of ZTC-PU 10% was 246 °C. The gradual increment in temperature was observed for ZTC-PU 20% (262 °C) and ZTC-PU 30% (286 °C). The 5% wt loss temperature of sample got steadily increased from 138 to 230 °C of ZTC-PU10% to ZTC-PU 30%, in the 20% wt loss also there was steady incline in temperature of ZTC-PU 10% (294 °C), ZTC-PU 20% (300 °C) and ZTC-PU 30% (306 °C) because of increased PDMS content in the zwitterionic polymer. This behaviour is possibly due to increased Si–O bond character which leads to higher thermal stability. These results give good agreement with the thermal stability of PDMS in the coatings [44]. Whereas, 50% wt loss and the samples remain at 400 °C showed ZTC-PU 10% and ZTC-PU 20% is increasing whereas ZTC-PU 30% is decreasing due to the formation of small voids and rough surfaces in the coating because of higher SBMA concentrations in the flanking blocks that creates associative character with polyurethane as a result forms chain contraction and phase separation [45]. This can be seen in SEM image (Fig. 8). Further, weight loss observed in the thermogram of ZTC-PU 30% (Fig. 6) between 100 and 200 °C might be attributed to the evaporation of water molecule physically absorbed by the coating. Hence, highest content of zwitterionic moiety present in the ZTC-PU 30% sample leads to absorbing the water molecule [46]. The DSC result (Fig. 7) also suggest that the glass transition temperature (T_g) of ZTC-PU

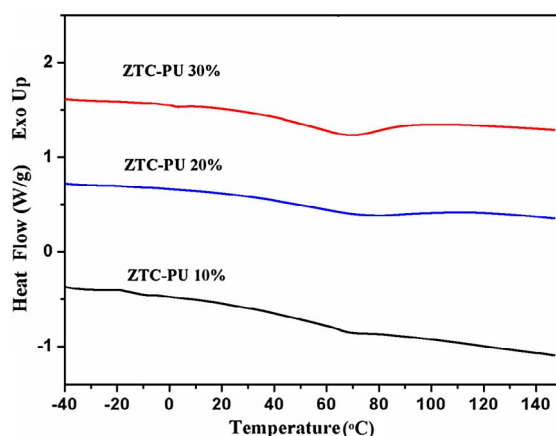


Fig. 7. DSC profile of zwitterionic triblockcopolymer-polyurethane (ZTC-PU).

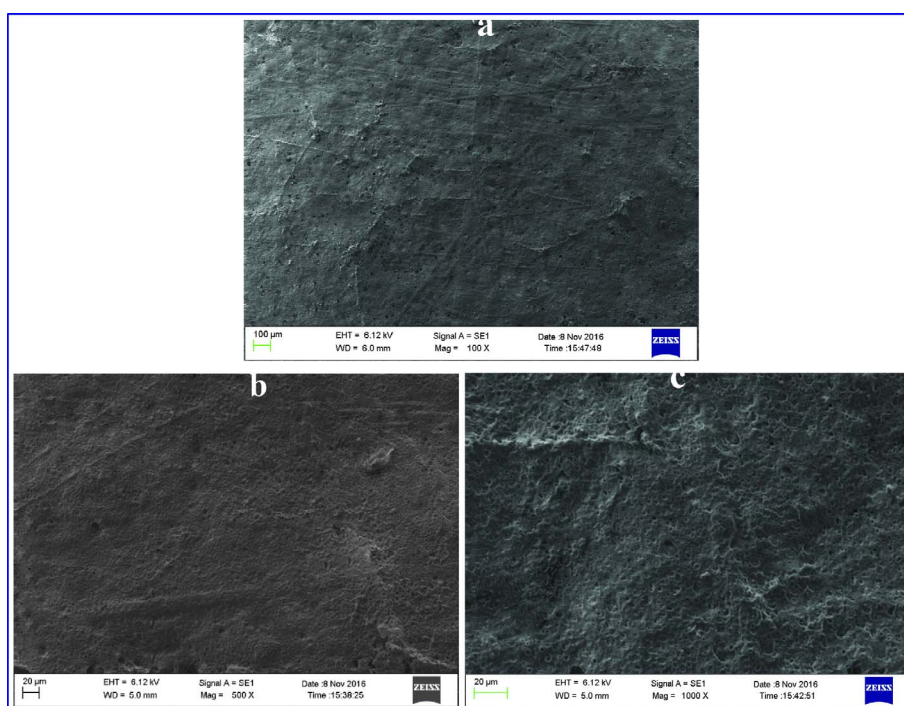


Fig. 8. SEM image of zwitterionic triblockcopolymer-polyurethane (ZTC-PU) (a) ZTC-PU 10%, (b) ZTC-PU 20% and (c) ZTC-PU 30%.

coatings shows T_g 63.88 °C, 67.72 °C and 72.52 °C of ZTC-PU 10%, ZTC-PU 20% and ZTC 30% respectively, the glass transition temperature (T_g) (Table 2) increase with increasing zwitterionic triblock copolymer. The reason might be assumed that the mobility of chain is restricted by zwitterionic group present in the polymeric chain. Further these groups enhance the polar-polar interaction and supports the formation of inter chain association, and also could be increase crystalline behaviour of the polymer. From the DSC curve (Fig. 7), the sharp T_g curve was observed with increased ZTC content in polyurethane. The effect of increasing T_g in DSC result gives good agreement with TGA onset degradation temperature (Table 1).

Table 2

Glass transition temperature (T_g) and water contact angle (θ) of zwitterionic triblockcopolymer-polyurethane (ZTC-PU).

Sample	T_g (°C)	Contact angle (θ)
ZTC-PU 10%	63.88	85
ZTC-PU 20%	67.72	98
ZTC-PU 30%	72.52	105

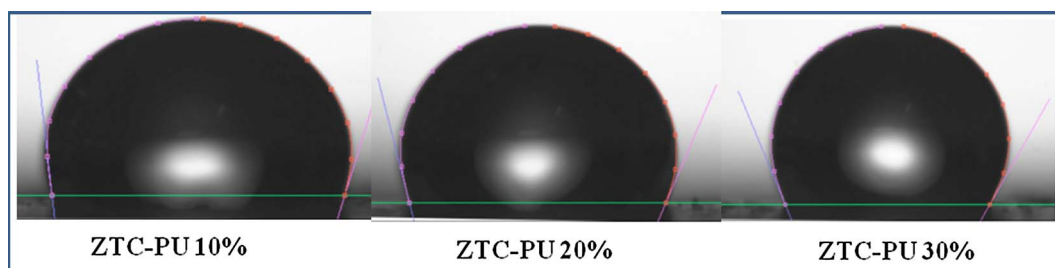


Fig. 9. Water contact angle image of zwitterionic triblockcopolymer-polyurethane (ZTC-PU).

3.5. SEM image and water contact angle measurement of ZTC-PU coatings

SEM micrograph (ZTC-PU 10%, ZTC-PU 20% and ZTC-PU 30% are shown in Fig. 8) shows that with increase in incorporation of ZTC in PU increase surfaces roughness and creates tiny voids. This is due to the large difference in the solubility parameter between polar Zwitterionic triblock copolymer and non-polar polyurethane. The molecular polarity causes internal stress in the system, to minimize the stress the low surface energy PDMS containing triblock copolymer migrates to the surface and shows the self-stratification behaviour [27,44]. The water contact angle (Fig. 9) of ZTC-PU 10%, ZTC-PU 20% and ZTC-PU 30% coatings are 85, 98 and 105° respectively (Table 2), this is due to the increasing PDMS content in the centre block. Also these results are in tandem with the SEM results (Fig. 8). There is an increase in the surface roughness and an increase in the contact angle was observed.

3.6. Anti-bacterial assay and marine bacterial inhibition test

The antibacterial activities of ZTC-PU films were tested against *Staphylococcus aureus* and *Escherichia coli*, *Pseudomonas aeruginosa*. Antibacterial activity measured based on the formation of inhibition zone loss of bacterial beneath and surrounding of the film placed on nutrient broth agar medium (Fig. 10(1)). The ZTC-PU films against *Staphylococcus aureus* showed good activities on ZTC-PU 20%, ZTC-PU 30%. Whereas, ZTC-PU 10% did not show any inhibition zone around the film also did not show any colour change or degradation by bacterial attack, at this concentration. Whereas, the same film showed antibacterial properties against gram negative species like *E. Coli* and *P. Aeruginosa* at 10 and 20% film concentration. Moreover, ZTC-PU 30% did not show any zone of inhibition against these strains. All these findings indicate that the percentage of zwitterionic triblock copolymer loading and the hydrophobic PDMS block in the polyurethane increase the antibacterial activity [47]. In addition, higher content of zwitterionic group lead to increase in the ionic character and increases the antibacterial efficacy [48,49]. The result suggested that the ZTC-PU 20% coating presented the best antibacterial activity among others.

The non specific marine organism was isolated and cultured from sea water. The isolate were proliferated into (a) lower concentration and (b) higher concentration (Fig. 10(2)). Antibacterial activity was measured based on the formation of inhibition zone at around the film placed in the bacterial culture. Inhibition zone were only observed in ZTC-PU 20% against these non-specific marine bacteria at both concentrations, whereas ZTC-PU 10% and ZTC-PU 30% did not show any zone of inhibition when tested against these isolates. In both the studies ZTC-PU 20% coating showed excellent activity against specific bacterial strains as well as against non specific marine isolate. The anti- bacterial properties of ZTC-PU coating attributed by active formation of electrostatic induced hydration of zwitterionic moiety [50]. In addition, the charge density, molecular weight and high fraction of zwitterionic triblock copolymer also plays a key role to inhibit the protein and bacterial adhesion [51].

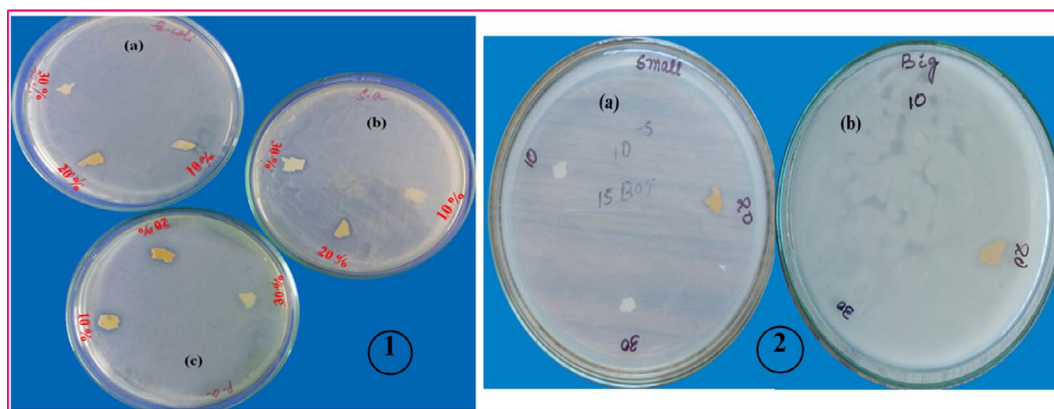


Fig. 10. (1) Antibacterial activity of ZTC-PU against specific bacterial strains (a) *E. coli*, (b) *S. aureus* and (c) *P. Aeruginosa*. (2) Non-specific marine bacterial activity of ZTC-PU films. (a) Lower concentration, (b) higher concentration.

4. Conclusion

In this study, an attempt has been made to develop zwitterionic triblock copolymer (ZTC) Poly(SBMA)-block-PDMS-block-poly (SBMA) – polyurethane(PU) ZTC-PU coatings for marine applications. The zwitterionic triblock copolymer was synthesized with higher concentration of SBMA in the flanking block through ATRP method. It was further formulated at different weight percentages (10, 20 and 30) with the silane functional acrylic polyol and H12 MDI for the final ZTC-PU coatings. In this attempt, we were able to successfully incorporate up to 30 wt% of zwitterionic triblock copolymer in the polyurethane system. This is usually difficult due to the large difference in the solubility parameter between polyurethane and zwitterionic triblock copolymer. Moreover, the simple synthesis of silane functionalized acrylic polyol was employed in the coating formulation. The amine group present in the silane moiety enhances the molecular polarity between polyurethane and water soluble zwitterionic triblock copolymer, which also endows proper phase mixing and uniform coating formation. TGA and DSC results show that the coatings are thermally stable. Water contact angle indicates that the coatings are hydrophobic in nature. Further, the coatings showed significant non-specific marine bacterial inhibition properties as well as good stability against gram positive *S. aureus* and gram negative *E. coli*, *P. aeruginosa*. These synthetic methods have paved way for the incorporation of higher amount of zwitterionic triblock copolymer into the polyurethane coatings. Hence, the present work provides effective and promising way to prepare stable zwitterionic-polyurethane coatings for non-specific marine bacterial inhibition.

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